TANZANIA HEADS OF ISLAMIC SCHOOL COUNCIL

FORM IV INTER ISLAMIC MOCK EXAMINATION 2024

CHEMISTRY 1

MARKING SCHEME

SECTION A (16 Marks)

1.

(i)	(ii)	(iii)	(iv)	(v)	(vi)	(vii)	(viii)	(ix)	(x)
C	B	D	B	B	B	E	C	D	

01 mark @ = 10 Marks

2.

COLUMN A	(i)	(ii)	(iii)	(iv)	(v)	(vi)
COLUMN B	H	E	I	D	B	G

01 mark @ = 06 marks

SECTION B (54 Marks)

3. (a) (i) Water **(01 mark)**

(ii) - Pure water is neutral room temperature its neither acidic nor basic.

- Cold water react with some metals to form metal hydroxides and liberate hydrogen gas.

- Steam can react with same metals to give the respective metal oxides and hydrogen gas. **03 marks @ 01 mark**

(iii) White anhydrous Copper (II) sulphate exposed to substance G (water) it dissolves to give a blue solution. **01mark**

(b) (i) Firewood and charcoal are cheap and most available but not recommended for domestic uses.

Reasons

- They are difficult to transport, handle and store.
- They pollute the environment during their production storage and use.
- They have low pyrometric burning effect.
- They have high content on non-combustible materials. **02marks@1 mark**

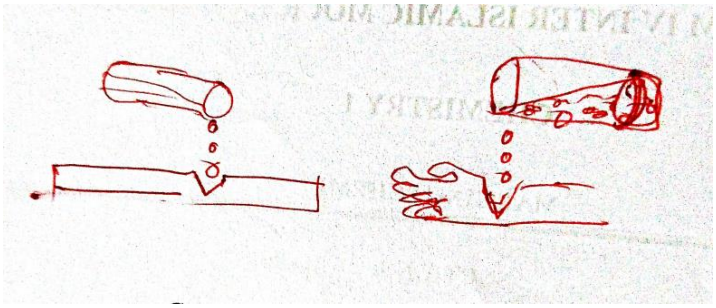
Any two points

(ii) This is because petrol has lower ignition points than kerosene which may cause fire hazards.

wazaelimu.com

Kerosene has high ignition point than petrol **[02marks]**

4. (a) Container A



Container B



Toxic

Container C



(b) Flammable **03 marks @ 01 mark**

Tetrahedral in shape (Diamond)	**Hexagonal in shape (Graphited)**
Has higher density 3.53g/cm^3	Has lower density 2.26g/cm^3
Bad conductor of heat and electricity	Good conductor of heat and electricity
Has higher refractive index is hardest	Has poor refractive index

It is hardest	It is soft and greasy
It melts at 4027°C and boil at 4830°C	It melts at 3600°C and boil at 4827°C

06 Marks @ 1 ½ marks. Any four points

5. (a) (i) Because cold foods have low kinetic energy which do not allow them to diffuse quickly in the air and can not travel a distance of several metres.

(ii) Because metal are good conductor of heat.

(iii) Because on heating, the temperature of the solution will increase which results in faster movement of particles and create space in water molecules so that more sugar gets dissolved./ temperature affect solubility

(iv) Because burning materials for class B and C fires are less denser than water and will float on top. Therefore, water instead of extinguishing them cause them to spread.

(v) Because they can be made to flow or move. Their molecules ran more from one place to another. **05 marks**

(b) (i) Painting

- This is the coating of substances such as metals with a special pigment or paint.

(ii) Oiling/Greasing

- This involves coating iron with oil. Some machine parts can be protected from rusting by oil regularly.

(iii) Galvanization

- This is the process of coating materials or steel with zinc. Example Galvanizing iron sheets.

(iv) Tin plating / electroplants

- This is the coating of iron material with tin.

(v) Use of silica gel

- Silica gel – Is a substance in the form of grains, at it absorbs moisture.

(vi) Sacrificial anode

- In this method, a more reactive metal than iron is attached to the protected material and will be consumed in favour at it. Therefore, it requires regular replacement for the protection to be.

01 @ any four points = 04 marks

6. (a) Data

Atomic number (Z) = 48

Mass number (A) = 112

Number of neuron (n) =

Number of electrode (e)

(i) Number of neutron

From A = z+n

112 = 48+n

112 – 48 = n

n = 64

∴ The number of neutron (n) = 64.

(ii) Structure of atom to represent in the nucleus of cadmium .



(iii) The nuclide notion of Cadmium

$^{112}_{48}Ca$ **06 marks**

(b) Rugiatu will finish cooking first. **01mark**

Because Rugiatu cut potatoes into small pieces which increased the surface area of the cooking vessel. When the surface is increased the rate of reaction also increased, so Rugiatu will finish cooking first. **02 marks**

7. (a) (i) Data given

Element 12M, E:C = 2:8:2, valency = 2

Element 17N, E:C = 2:8:7, valency = 8-7=1

The chemical formula of the compound formed by combination M and N.

(1) Element: M N T

(2) Valence its ion M N

(3) Exchange of valency M^2 N^1 M_1N_2

The compound formed is MN_2. **01mark**

(ii) Ionic/Electrovalent bond. **01 mark**

(iii) Characteristics of ionic/Electrovalent bond.

- They are generally soluble in water.
- They conduct electricity solution or molten form but not in their solid forms.
- They are usually crystalline solids at room temperature
- They are generally insoluble in non-polar solvents such as hexane.

03 marks @ 01 mark Any three points

(b) (i) But-i-ene

(ii) 2,2-dimethy/butance

(iii) 2,3-dimetybjtane

(iv) 2-chloro-3-methylbutane **04marks @ 01 mark**

8. (a) Data given

Mass of base M_2CO_3 = 17.25g

Volume of the solution = 250cm^3 = 0.25dm^3

Concentration of base = $\frac{mass\ (g)}{volume\ (dm^3)} = \frac{17.25g}{0.22dm^3} = 69g/dm^3$

Molarity of acid (Ma) = 0.8m

Volume of acid (Va) = 25cm^3.

Volume of base (Vb) = 20cm^3

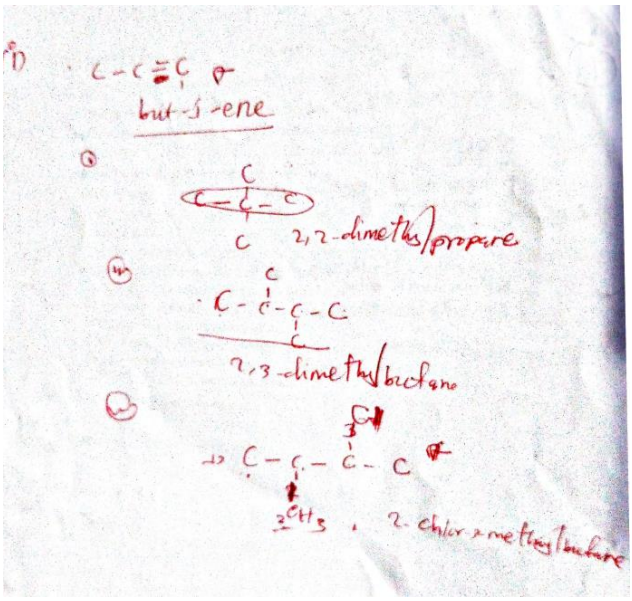
From balanced chemical equation

$2HCl_{(aq)} + M_2CO_3(aq) \rightarrow 2MCl(aq) + Cl_2O_{(l)} + CO_2(g)$

Number of moles of acid (na) =2

Number of moles of base (nb) = 1

(i) From titration law



∴ The relative atomic mass of m is 39.

(ii) Element m is potassium (k) **01mark**

(b) (i) Neutralization reaction. **01 mark**

(ii) Application of neutralization reaction in our daily life.

- Relieving indigestion – Excess of hydrochloric acid produced in our stomach which can be neutralized by antacid (magnesium or sodium) hydrogen carbonate.
- Soil treatment – Living material (quicklime and calcium hydroxide as antacid (Gase) is added to acidic soil to neutralize them.
- Treating factory wastes – Liquid waste from factories often contain acids and base, acidic wastes can be controlled by adding bases such as calcium hydroxide to neutralize them.
- Preventing formation of acidic rain – Acidic rain can be reduced by putting air pollution devices containing bases are fitted pipes and chimney to neutralize the acidic compound before reaching the atmosphere.
- Neutralizing accidental spills – Anacid or an alkali spills on the floor or work surface in the laboratory can be neutralized.
- Manufacture of fertilizer, fertilizer like Ammonium sulphate can be produced by neutralization reaction between sulphuric acid and ammonium hydroxide.

03 marks @ 1 mark any three points

9. Ways through which soil fertility may be lost:

(i) Leaching
(ii) Soil capping
(iii) Soil erosion
(iv) Monocropping
(v) Change in soil pH
(vi) Burning of vegetation
(vii) Accumulation of salts

Any 6 points @ 02 marks = 12 marks
Introduction 2 marks
Conclusion 1 mark
TOTAL 15 MARKS

10. Importance of changes of state of matter.

(i) Water cycle
(ii) Refrigeration
(iii) Drying of materials
(iv) Working of steam engines
(v) In refinery
(vi) In extraction an purification of metals (metallurgy)

Any 6 points @ 2 marks = 12
Introduction 2 marks

Conclusion 1 mark
TOTAL 15 Marks

11. Causes of noise pollution:

(i) Road traffic noise
(ii) Construction activities
(iii) Air traffic noise
(iv) Work place noise
(v) Residential noise
(vi) Events involving fire works
(vii) Commercial noise
(viii) Industrial noise

Any 6 points @ 02 = 12 marks
Introduction = 02 marks
Conclusion = 01 mark

TOTAL 15 Marks

Wabillah Tawfiq